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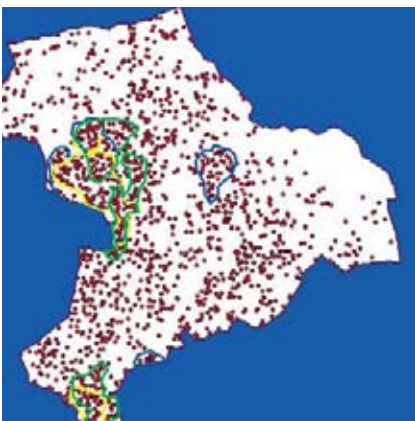


## Mapping Crime: Understanding Hot Spots

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## Chapter 2. Methods and Techniques for Understanding Crime Hot Spots

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This chapter presents a number of methods and techniques to understand and describe patterns of hot spots in crime data. It explains the advantages and disadvantages of certain techniques, focusing on methods that are easy to understand and practical to apply. Four data sets are used to test the methods. The results help evaluate how the methods improve understanding of crime patterns. This chapter does not attempt to find the *optimal* method. Rather, it presents a procedure for applying a number of complementary methods that can help analysts understand hot spots in the data.

The four data sets are from the London Metropolitan Police Force's Crime Report Information System for Hackney Borough Police for the period June 1999 through August 1999. This chapter contains tests developed and conducted to help analysts understand hot spots in representative samples of crime data sets of—

- All crime (9,972 records) and three subsets.
- Street robbery (588 records).
- Residential burglary (1,068 records).
- Vehicle crime (1,747 records).

The geographic area for these data is insignificant to explain the different methods for trying to understand crime hot spots. The data should be treated merely as point events of crime within a geographical boundary area. In some cases,

these points will be aggregated to small area geographies such as beats or census blocks to demonstrate the applicability of certain techniques. The methods discussed and tested on these data are perfectly applicable to analysts' own data.

### Preliminary global statistical tests

A number of simple-to-use global statistical tests can be used to help analysts understand general patterns in the crime data presented here. These tests include—

- Mean center.
- Standard deviation distance.
- Standard deviation ellipse.
- Tests for clustering.

### Mean center

The mean center point can be used as a relative measure to compare spatial distributions between different crime types or against the same crime type for different periods of time (i.e., for measuring spatial shifts in the same crime type). For example, exhibit 1 shows the mean center points for all crime, street robbery, residential burglary, and vehicle crime. The all-crime mean center can be used as a control to compare against crime type subsets. Residential burglary has a mean center that is more north than all crime

and the two other crime subsets. The mean center for vehicle crime is the farthest south of all the crime types, and street robbery is nearly as far north as that for residential burglary but slightly farther west. These mean centers can be used to generally indicate that residential burglary and robbery offenses show a greater tendency to occur in the northern part of the borough and that vehicle crime affects the southern areas of the borough more.

### Standard deviation distance

Measures of standard deviation distance help explain the level and alignment of dispersion in the crime data. These statistics are best used as relative measures, comparing crime types against each other or the same crime types for different periods of time. Exhibit 2 shows the standard deviation distances for the four crime types. The greater the standard deviation dis-

tance, the more dispersed are the crime data. These results show that vehicle crime is the most dispersed; robbery is the least.

### Standard deviation ellipses

Levels of dispersion also can be presented using standard deviation ellipses. The size and shape of the ellipse help explain the degree of dispersion, and its alignment helps to explain the crime type's orientation. Exhibit 3 shows standard deviation ellipses for the four crime types. The subtle differences between the ellipses help describe the relative differences in dispersion and alignment of the four crime types' patterns. The ellipse with the smallest area (robbery) is the least dispersed of the crime types. The position of the robbery ellipse farther north of all crime and vehicle crime, but slightly farther south of the residential burglary ellipse, reflects its

**Exhibit 1. Mean center points for the four crime categories**

- All crime
- Street robbery
- Residential burglary
- Vehicle crime
- Borough boundary



**Exhibit 2. Standard deviation distance results for the four crime categories**

| Crime type           | Standard deviation distance (meters) |
|----------------------|--------------------------------------|
| All crime            | 1,807.94                             |
| Robbery              | 1,749.94                             |
| Residential burglary | 1,806.28                             |
| Vehicle crime        | 1,820.85                             |

mean center. Its north-west, south-east orientation also helps to describe the general direction toward which robbery crimes have a tendency to be patterned.

### Tests for clustering

The fourth and probably most useful of the preliminary global statistics are those that test for clustering. Crime analysts often

assume that crime distributions are clustered, and whether clusters exist or not, some can be identified from random crime distributions. Testing for clustering is the first step in revealing whether data has hot spots of crime.

Several approaches can be applied to test for clustering in crime distributions. Most methods incorporate the basic principles of hypothesis testing and classical statistics, in which the initial assumption is that the crime distribution is one of complete spatial randomness (CSR). By setting the CSR assumption as the null hypothesis, the crime distribution can be compared against a set significance level to accept or reject the null hypothesis. Some tests for clustering are the nearest neighbor index and the spatial auto correlation tests.

**Nearest neighbor index (NNI).** The NNI is a simple and quick method to test for

**Exhibit 3. Standard deviation ellipses for the four crime categories**

- All crime
- Robbery
- Residential burglary
- Vehicle crime



evidence of clustering. The NNI test compares the actual distribution of crime data against a data set of the same sample size, with random distribution. It can be applied when the user has access to data in which each point relates to individual crime events (irrespective of whether some of these events are mapped on top of each other at exactly the same location). The NNI method is explained in the following steps:

- For each point, calculate the distance to the nearest neighbor point.
- Sum the nearest neighbor distance for all points and divide by the number of points in the data. This value is the observed average nearest neighbor distance.
- Create a random distribution of the same number of crime points covering the same geographic area, and for each point calculate the distance to each nearest neighbor point.
- Calculate the sum of the nearest neighbor distances for all these randomly distributed points and divide by the number of points in the data. This value is the average random nearest neighbor distance.
- The NNI is then the ratio of the observed average nearest neighbor distance against the average random nearest neighbor distance.

If the result generated from the NNI test is 1, then the crime data are randomly distributed. If the NNI result is less than 1, then the crime data show evidence of clustering. An NNI result that is greater than 1 reveals evidence of a uniform pattern in crime data.

Exhibit 4 shows the NNI results for the four crime data sets. All four sets show evidence of clustering in their distribution.

The results also show the differences between the NNI results for a minimum bounding rectangle area and the actual catchment area of the crime points. When the actual catchment area is known or can easily be calculated, it should be used in the calculation of the NNI. If the area is not known, a minimum bounding rectangle around the crime distribution often is used to calculate the area representing the crime data's catchment zone.

However, minimum bounding rectangle areas used for NNI tests often can create misleading results for describing point distributions. To test and show evidence of this, analysts purposely designed a regular distribution of points and applied three different types of bounding areas: minimum bounding rectangle, bounding convex hull, and true boundary area. After applying NNI tests, the bounding convex hull method and true boundary area revealed similar results, correctly describing the point

**Exhibit 4. Nearest neighbor analysis results for the four crime data sets**

| Crime type and bounding area* | NNI  | Z-score |
|-------------------------------|------|---------|
| <b>All crime</b>              |      |         |
| Bounding rectangle area       | 0.32 | -129.11 |
| True boundary area            | 0.46 | -103.14 |
| <b>Robbery</b>                |      |         |
| Bounding rectangle area       | 0.59 | -19.14  |
| True boundary area            | 0.80 | -9.20   |
| <b>Residential burglary</b>   |      |         |
| Bounding rectangle area       | 0.57 | -27.14  |
| True boundary area            | 0.74 | -16.46  |
| <b>Vehicle crime</b>          |      |         |
| Bounding rectangle area       | 0.52 | -38.73  |
| True boundary area            | 0.72 | -22.16  |

\*Crime distribution is clustered for all areas.

data's distribution to be regular. The result from the minimum bounding area suggested the same distribution to be random. Where the true boundary area is not available, a convex hull that bounds the limits of the point distribution will return a more accurate result for evidence of clustering than a minimum bounding rectangle.

A z-score test statistic can be applied to help place confidence in the NNI result. This test for statistical significance describes how different the actual average nearest neighbor distance is from the average random nearest neighbor distance. The significance of the z-score can be found in any table of standard normal deviations. The general rule to follow is that the more negative the z-score, the more confidence can be placed in the NNI result, bearing in mind that for smaller sample sizes the z-score will be less than that for larger samples of crime points.

**Test for spatial auto correlation.** Spatial autocorrelation techniques test whether the distributions of point events are related to each other. Positive spatial autocorrelation is said to exist where events are clustered or where events that are close together have similar values than those that are farther apart.

Spatial autocorrelation tests, of which Moran's I is one commonly applied method, have been used previously to test for evidence of crime event clustering (Chakravorty, 1995). Spatial autocorrelation techniques require an intensity value, be it a weighting linked to the event or a count of crimes where the crime point relates to the coordinate of an area to which crime events have been aggregated (e.g., the centroid of the area). If the original crime event data exist as accurate point georeferenced data, aggregating this data to a common point will lose spatial detail. With the increased availability of accurate and precise geocoded records of crime, it

would seem more important to use methods that do not require an intensity value but retain and perform tests on the original crime event point data.

Spatial autocorrelation methods include the following:

- **Moran's I.** If analysts have access only to crime point data that are aggregate counts (representing the number of crime events within a certain geographic area, e.g., census blocks), an appropriate method to apply to test for clustering is the spatial autocorrelation technique, Moran's I. (See exhibit 5.) Moran's I statistic works by comparing the value at any one location with the value at all other locations (Levine, 2002; Bailey and Gatrell, 1995; Anselin, 1992; Ebdon, 1985). Moran's I requires an intensity value for the crime point, which is often represented as the centroid of the geographic boundary area. This point is then assigned an intensity value. For crime applications, this most often is the count of crimes within that geographic area. The Moran's I result varies between -1.0 and +1.0. Where points that are close together have similar values, the Moran's I result is high. The significance of the result can be tested against a theoretical distribution (one that is normally distributed) by dividing by its theoretical standard deviation. (For more details on this test, see Levine, 2002.)
- **Geary's C statistic.** A second spatial autocorrelation method is Geary's C statistic. (See exhibit 5.) This method is best applied to describe differences in small neighborhoods. (Moran's I gives more of a global indicator of spatial relationships; Levine, 2002). Geary's C statistic is a measure of the deviations in intensity values of each point with one another. The values of C typically vary between 0 and 2, where values less than 1 indicate evidence of positive



spatial autocorrelation and values greater than 1 indicate evidence of negative spatial autocorrelation. Similar to Moran's I, the Geary coefficient can be tested for significance against a theoretical distribution (one that is normally distributed) by dividing by its theoretical standard deviation.

The differences in the sensitivities of the two spatial autocorrelation tests are noted in the results on the four crime categories. For example, evidence exists of positive spatial autocorrelation for robbery at a more global level. However, the Geary coefficient reveals that at the smaller neighborhood level, areas that have a high number of robberies are surrounded by areas with a low number of robberies.

The preliminary global statistics reveal a wealth of knowledge about the crime data even before the mapped distributions are explored in detail. The analysts have generated an understanding of the global patterns in the crime data and shown that evidence of clustering exists in all of the four crime categories. This clustering does, however, vary at different scales. The dispersion tests have revealed that although clustering exists in the data, the crime hot spots for vehicle crime are more dispersed than any of the other crime categories. These tests, therefore, begin to reveal a picture of what the hot spot maps may look like. For example, when the robbery data are mapped, analysts can expect hot spots of crime to exist and to be con-

centrated together more than any other crime category.

The following sections explore various techniques for mapping crime. These sections mainly use the vehicle crime data set to explore the different applications of these techniques to help in understanding hot spots of crime.

## Crime mapping techniques

### Point mapping

The most common approach for displaying geographic patterns of crime is point mapping (Jefferis, 1999). Point mapping is popular mainly because it is a simple digital version of a familiar and traditional method of placing pins representing crime events onto a wall map. In a digital application, if these individual geographic point objects are suitably attributed with information, such as the code describing the type, date, and time of offense, sets of points meeting particular conditions can be simply and quickly selected. These selections can then be displayed using suitable symbology representing the crime category displayed. However, trying to interpret spatial patterns and hot spots in the crime point data can be difficult, particularly if the data sets are large.

Exhibit 6 shows the 1,747 events of vehicle crime. The large volume of points shown on map A makes it difficult to feel

**Exhibit 5. Spatial autocorrelation results for the four crime categories**

| Crime type           | Moran's I | Z-score  | Geary's C | Z-score    |
|----------------------|-----------|----------|-----------|------------|
| All crime            | 0.003116  | 2.161428 | 0.979587  | -1.635861  |
| Robbery              | 0.000660  | 1.185983 | 1.022821  | 3.585178   |
| Residential burglary | 0.002996  | 2.113746 | 0.987718  | -1.929432  |
| Vehicle crime        | 0.006658  | 3.568309 | 0.918423  | -12.815473 |

completely confident in identifying the hot spots of this particular crime. The preliminary global tests revealed that clusters of vehicle crime were evident, but that these hot spots were more dispersed than any of the other crime types. To demonstrate the ambiguity in trying to identify the hot spots of crime from this particular data set, the analysts asked three crime analysts who were not familiar with the study area to draw the location of the top three hot spots of vehicle crime on map B. The three analysts identified two common areas as being hot spots of crime, but the areas drawn at these locations differed in size and shape. Two hot spots identified by analyst 1 are completely different from those identified by analysts 2 and 3.

Which analyst is correct? At this stage it is difficult to tell because each hot spot drawn is plausible. Not all of those drawn will be completely correct. This example

demonstrates the difficulty in trying to consistently identify crime hot spots from point data. Also, at certain locations, what appears to be a single crime point may be more than one crime point. These points at coincident locations are impossible to identify using the point map in exhibit 6. Point maps do have their application for mapping individual events of crime, small volumes of crime, and repeat locations through the use of graduating symbol sizes (see exhibit 7), but they can become less effective for identifying hot spots of crime, particularly from large data volumes.

### Spatial ellipses

The application of spatial ellipses for attempting to identify crime hot spots has a long tradition in crime mapping. The Spatial and Temporal Analysis of Crime (STAC) software distributed by the Illinois Criminal Justice Information

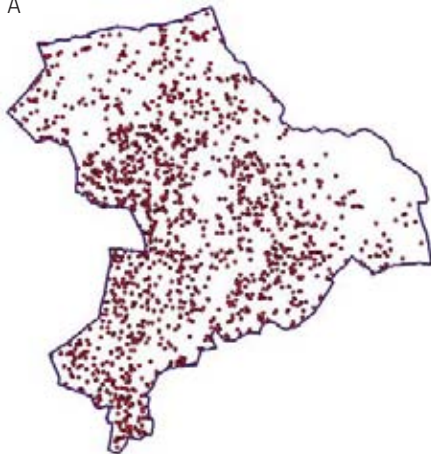
#### Exhibit 6. A: Point map of vehicle crime

##### B: Three sets of hot spots identified by different analysts

- Crime analyst 1
- Crime analyst 2
- Crime analyst 3

Point hot spots

A



B





Authority is one type of available software used for spatial ellipses. The Space Analyzer component of STAC works by creating standard deviational ellipses around crime point clusters. Other spatial ellipse techniques include hierarchical clustering and the K-means clustering routine.

**Hierarchical clustering.** This method uses a nearest neighbor analysis technique to identify groups of a minimum number of user-defined points. The nearest neighbor analysis technique used identifies only those points that are closer than expected under spatial randomness. The first set of ellipses generated through this process is referred to as first-order clusters. Grouping the first-order clusters can then generate second-order ellipses. This process can then be repeated until all crime points fall

into a single cluster or when the grouping criteria fails (Levine, 2002).

**K-means clustering.** The K-means technique creates a user-defined number (K) of ellipses by partitioning the crime point data into groupings. The routine finds the best positioning of the K centers and then assigns each point to the center that is nearest. Exhibit 8 shows five ellipses created using the K-means method. The method demonstrates how spatial ellipse techniques are useful for grouping crime point clusters and revealing areas for closer inspection. However, the ellipse outputs also demonstrate certain weaknesses in these types of techniques if analysts are trying to accurately identify the location of crime hot spots. Crime hot spots do not naturally form spatial ellipses. These

**Exhibit 7. Properties with multiple breakins (January 1996–July 1997)**



methods, therefore, do not represent the actual spatial distribution of crime and can often mislead the analyst to focus on areas of low crime importance within an ellipse. Also, all of these techniques require a thorough understanding of the routines at work because each method requires a number of parameters to be entered. Users with a vague understanding of the actual methods are given few rules to enter appropriate parameters, thus introducing ambiguity and influencing variability in the final output. For example, different crime analysts investigating crime hot spots from the same data may produce different results because of the different parameters they have chosen to enter into the routine. Grouping events into elliptical clusters also excludes from any visual result those events that do not belong to a cluster.

Four of the five ellipses drawn using the K-means method partially overlap the hot spot areas drawn by the three crime analysts from the point map. One ellipse differs completely from an area identified by analyst 1. At this stage it is difficult to understand which one is correct or which

method returns more accurate results. Both methods are plausible and importantly present the opportunity to explore the nature of crime in these areas in more detail. However, neither method presents the immediate opportunity to prioritize the main crime hot spots to assist in prevention targeting.

### Thematic mapping of geographic boundaries

A popular technique for representing any spatial distribution is geographic boundary thematic mapping. These geographic boundaries usually are defined administrative or political areas such as beats, census blocks, polling districts, wards, or borough boundaries. Crime events mapped as points can be aggregated to these geographic region areas. These counts of events by their geographic areas can then be thematically mapped to display the spatial pattern of crime across the area of interest (see exhibit 9).

When thematic maps of this type are created, the user is required to identify the type of range to represent the distribution

**Exhibit 8. K-means elliptical clusters for vehicle crime**



of crime (e.g., equal count, equal ranges, natural break, standard deviation, quantile, or custom range; for a good review of these different types of range settings, see Harries, 1999). This freedom to choose a range of any type creates variability in any thematic map produced for identifying crime hot spots. Different range types structure the groupings of geographic boundary crime counts into differing threshold categories. At what range threshold can the crime analyst confidently conclude that geographic boundaries in that range and above represent hot spots of crime? In other words, when is a hot spot a hot spot?

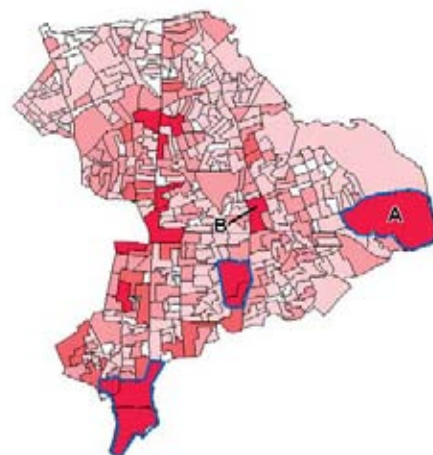
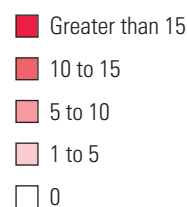
Considering the theory and application behind the map the user is trying to produce and the target audience is important at this stage. If the application is to identify the hot spots of crime, the range thresholds should be structured to focus on revealing the locations of these high-volume crime areas. The analysts also should consider structuring the thematic ranges in a manner that is easy for the audience to understand. The map should be the central message and the thematic ranges should follow in a logical sequence. If the thematic ranges require explanation or confuse the audience, the analysts have lost the opportunity to present their central message: mapping and displaying crime hot spots. Mapping is an iterative process. The first map a user produces is unlikely to be the one printed or included in a report. Decide on the message and follow through with this message by testing different settings available for the thematic thresholds.

Exhibit 9 shows the vehicle crime data thematically mapped by census blocks. The range created is a custom one that follows in a logical and easy-to-understand sequence. It identifies those census blocks that are grouped in the highest crime count categories. From this map, a

crime analyst was asked to identify those areas he believed to be the three main hot spots of crime. These are the three hot spot areas drawn on the map. Notice how these interpreted hot spot areas differ from those drawn directly from the point data and those identified by spatial ellipses. Which map is correct?

Thematic maps such as the one in exhibit 9 tend to attract the audience to the largest areas shaded in the top threshold color range. Therefore, the single census block A has been selected as a hot spot as it stands out boldly. Census block B has not been selected as a hot spot, yet its area is one-eighth of that of census block A and it has a similar crime count, as it features in the top threshold range. This reveals a problem with using geographic boundary thematic maps to identify hot spots of crime. Due to the varying size and shape of the census blocks (and most

**Exhibit 9. Vehicle crimes mapped by census tract**



geographic boundaries), thematic shading can mislead the audience in identifying where the spatial cluster of crime may exist. Census block A may contain an area with a cluster of crime events. These crime events, however, may be evenly spread across the whole area of this large census block. Thus an actual hot spot in this area may not exist. If this was the case, the map in exhibit 9 may misinform any targeting initiative to hot spots of crime identified by this method.

An additional problem with mapping by defined boundaries is that presented by the Modifiable Area Unit Problem (MAUP) (Bailey and Gatrell, 1995; Openshaw, 1984), in which changes in the geographic boundary areas used to thematically represent the distribution of crime can affect and further mislead map interpretation. However, the geographic boundary thematic mapping method should not be completely discounted. Boundary thematic maps are important map outputs, as the areas they represent often are geographic regions used for political and administrative purposes. For example, a police inspector may have management responsibility for combating crime in a group of geographic areas. This inspector will want to be informed with general information such as the thematic map shown in exhibit 9. The inspector will also most likely be interested in receiving summarized information of patterns that exist in neighboring geographic areas (e.g., a geographic boundary thematic map, with a table of crime counts by categories for these areas of interest). These methods do, therefore, have an important application for providing summarized management information across areas of accountability, but may mislead focused crime prevention targeting because of failing to reveal patterns within and across the geographic division of boundary areas.

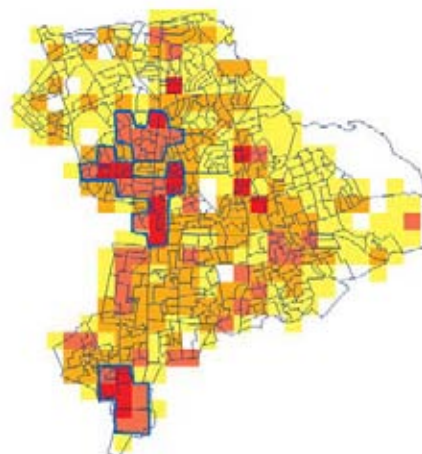
### Quadrat thematic mapping

To get over the problem of varying sizes and shapes of geographic administrative boundary areas, uniform grids (or quadrats) can be drawn across the study area and thematically shaded. The unit to thematically map could be a count of crimes per grid cell or a density value calculated from the count and cell area. If the grids are small enough, the analysts can then expose if hot spots of crime actually exist within large geographic boundary areas, while retaining and displaying high crime volumes in the smaller geographic areas. Exhibit 10 shows a 250-meter (m) quadrat thematic map of vehicle crime. Similar to the other mapping outputs, a crime analyst has drawn areas on the map that he interprets as being the three main hot spots (shown in blue). Interestingly, the quadrat hot spots identified fit closely

**Exhibit 10. 250-m quadrat thematic map**

- Greater than 15
- 10 to 15
- 5 to 10
- 1 to 5
- No crime

Vehicle crimes by 250-m quadrats



with a number of the hot spots identified from the point map. In addition, notice that the census block A identified as a hot spot in the geographic boundary thematic map (exhibit 9) is not revealed as a hot spot using this method. The crimes that contributed to the high count in this large census block are generally spread across the whole geographic region. The quadrat method, therefore, appears to offer a more accurate method for identifying hot spots of crime, particularly where applications for crime prevention targeting are required.

However, the quadrat method still restricts analysts to using a defined geographic area to represent the size and shape of crime hot spots. The restriction of using thematically shaded, defined geographic grid cells often results in loss of spatial detail within each quadrat and across quadrat boundaries. This can lead to problems of inaccurate interpretation. The map also looks blocky and does not entirely invoke interest. A common request to address this problem is to reduce the grid cell size. All this does is produce a speckled map of small, thematically shaded grid cells that fail to draw the user to any firm conclusions as to the size, shape, or location of crime hot spots for resource targeting. Also, issues with thematic range settings still exist with this method, including those that relate to the MAUP.

### Interpolation and continuous surface smoothing methods

**Interpolation.** Interpolation is an increasingly popular method for visualizing the distribution of crime and identifying hot spots. It aggregates points within a specified search radius and creates a smooth, continuous surface that represents the density or volume of crime events distributed across the area. Common interpolation techniques, such as inverse distance weighting, triangulation with smoothing,

kriging, and splining are all designed to use an intensity, population, or 'z' value taken from sample locations to estimate values for all locations between sample sites. For example, interpolation techniques are commonly used to create surfaces representing the distribution of rainfall, where the values between rain gauges are estimated from a function that considers the rainfall readings and the distribution of sample sites (i.e., rain gauges). With crime data, sample sites with an intensity value do not necessarily exist. Neither would it make sense to apply one of these techniques to estimate the number of crimes that may have occurred between the existing crime point locations. No crimes have been recorded in the areas between crime points, so the analysts should avoid methods that aim to create estimated intensity values in the gaps between the points. Instead, surfaces that the analysts wish to create that represent the distribution of crime should act as visualizations for helping them understand crime patterns. Methods that suit the analysts' application should therefore represent, as a continuous surface, the relationships or densities between crime point distributions.

**Quartic kernel density estimation.** The most suitable method for visualizing crime data as a continuous surface is quartic kernel density estimation (Chainey et al., 2002; McGuire and Williamson, 1999). The quartic kernel density method creates a smooth surface of the variation in the density of point events across an area. The method is explained in the following steps:

1. A fine grid is generated over the point distribution (see exhibit 11). In most cases, the user has the option to specify the grid cell size.
2. A moving three-dimensional function of a specified radius visits each cell and calculates weights for each point within the kernel's radius (see exhibit 12). Points

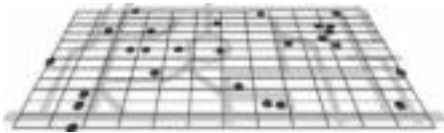
closer to the center receive a higher weight and therefore contribute more to the cell's total density value.

3. Final grid cell values are calculated by summing the values of all circle surfaces for each location (see exhibit 13).

The quartic kernel estimation method requires two parameters to be set prior to running. These are the grid cell size and bandwidth (search radius). Bandwidth is

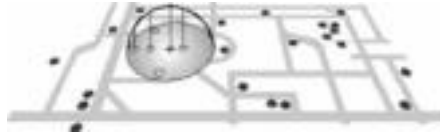
the parameter that will lead to most differences in output when varied. Guidelines exist for working out suitable values for these two parameters. For crime mapping applications, a suitable method to follow for choosing bandwidth is that suggested by Williamson et al. (1999), where the bandwidth relates to the mean nearest neighbor distance for different orders of K. These nearest neighbor distances for different orders of K usually are calculated as

**Exhibit 11. A fine grid is placed over the area covered by the crime points**



Source: Ratcliffe, 1999a.

**Exhibit 12. A search radius (or bandwidth) is then selected**

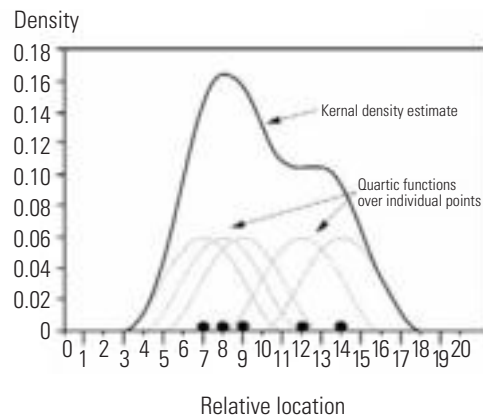


Source: Ratcliffe, 1999a.

Note: Within the bandwidth, intensity values for each point are calculated. Points are weighted, so that incidents closer to the center contribute a higher value to the cell's intensity value.

**Exhibit 13. Grid cell values are calculated**

Kernel Density Estimates  
Summing of Quartic Kernel Function



Source: Levine, 2002.

Note: Each grid cell value is the sum of values of all circle surfaces for each location.



part of the nearest neighbor analysis test for clustering. If this method is adopted, the user is still required to choose which K order to apply. This can be regarded as an advantage of the technique because users are prompted to think about the data they are mapping and apply the K order's mean nearest neighbor distance as the bandwidth relevant to their crime mapping application. For example, a smaller bandwidth would be used for an application that requires output for focused police patrolling than for one that requires a more strategic view of crime hot spots.

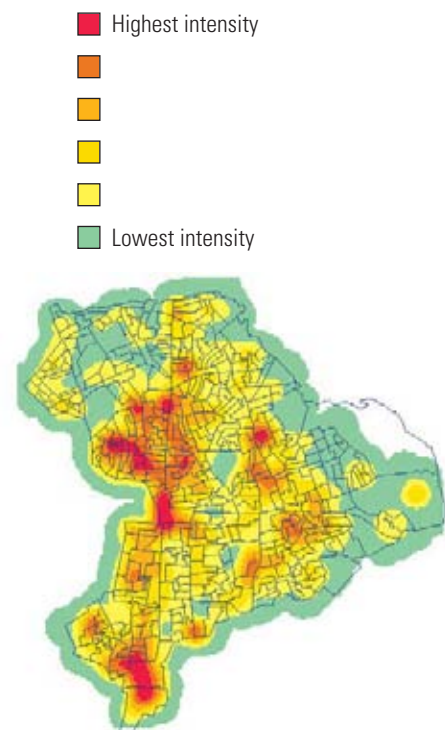
Users also are required to specify a grid cell size. This can be regarded as a positive feature in kernel estimation because users have the flexibility to set sizes relevant to the scale at which the output will be viewed. Large cell sizes will result in more coarse or blocky-looking maps but are fine for large-scale output, while smaller cell sizes add to the visual appeal of the continuous surface produced but may create large file sizes. When unsure of the cell size, the user can follow the methodology of Ratcliffe (1999b), who divides the shorter side of the minimum bounding rectangle (or the shortest of the two extents between minimum X/maximum X or minimum Y/maximum Y) by 150. Cell size for the area that the data cover has been set to 30 m. The cell values generated from quartic kernel estimation also are in meaningful units for describing crime distributions (e.g., the number of crimes per square kilometer). Exhibit 14 shows the quartic kernel density estimation surface generated from the vehicle crime point data, where the bandwidth chosen was a K order of 16 (220 m).

The quartic kernel density estimation method creates understandable grid cell value outputs that relate to crime and requires fewer parameters to be set than those required for many other methods. The parameters entered can relate to the

spatial distribution of the points being analyzed. The method also has the advantage of deriving crime density estimates based on calculations performed at all locations (Levine, 2002), and retains some practical flexibility in map design.

The increased application of this type of continuous surface smoothing method is due largely to its more common availability and visual appeal. Continuous surface hot spot maps allow for easier interpretation of crime clusters and reflect more accurately the location and spatial distribution of crime hot spots. The quartic kernel density estimation method also considers concentrations of crime at all event levels, rather than cluster grouping some and discounting others. As their appeal has increased, however, few questions are

**Exhibit 14. Quartic kernel density estimation surface for vehicle crime using a bandwidth of 220 m (K16)**



being asked of the outputs generated. The issue over which thematic range to choose to represent the different thematic thresholds remains a problem. Many agencies often fail to question this validity or statistical robustness of the map produced, being caught instead in the visual lure of their sophisticated looking geo-graphic. Thus, little regard is given to the legend thresholds that are set to help the analyst decide when a cluster of crimes can be defined as a hot spot. For example, the number of hot spots on a map showing the distribution of crime as a kernel density surface depends on the ranges selected by the map designer to show spatial concentrations of these point events. Careful selection of range settings is therefore required, but this opens the opportunity for crime analysts to create different kernel density estimation hot spot maps of crime from the same data.

One method that has been suggested to help standardize the thematic threshold settings of kernel density estimation hot spot maps is the application of incremental multiples of the grid cells' mean (Chainey et al., 2002). Calculations for the mean are applied only to grid cells that have a value of greater than 0 and that are within the study area boundary. From this grid cell set, the mean can easily be calculated within a geographic information system (GIS) with grid cell thematic thresholds set at—

- 0 to mean.
- Mean to 2 mean.
- 2 mean to 3 mean.
- 3 mean to 4 mean.
- 4 mean to 5 mean.
- Greater than 5 mean.

Exhibit 15 shows the application of this incremental mean threshold approach for

vehicle crime. The method is visually appealing and structures the thematic thresholds to clearly identify areas of highest crime concentration. It is simple to apply because it requires only the calculation of the grid cell mean; it uses K order mean nearest neighbor distances to define bandwidths; it retains flexibility in map design by allowing the user to apply different K order bandwidths and grid cell sizes to the suited application; and it uses a consistent methodology to separate thematic thresholds.

As a statistic, the mean is an easy value for novice map readers to understand. Increments of the mean would be more obviously linked to increasing values and their relative significance. This makes the incremental mean threshold approach immediately appealing as a method to define crime hot spot legend thresholds.

Exhibit 16 shows the results of applying this hot spot threshold approach, based on quartic kernel density estimation surfaces, to the robbery and residential burglary data. The incremental mean legend threshold approach consistently defines mapped crime hot spots. In addition, the analysts can now observe the global descriptions calculated earlier and see that they match the crime hot spot mapping output. The kernel surface hot spot maps for vehicle crime, robbery, and residential burglary display areas of crime clustering, and when compared against each other show relative levels of dispersal. For example, the hot spot maps of vehicle crime show a greater degree of dispersion compared to that of robbery.

## Local indicators of spatial association statistics

A more advanced method to help understand hot spots of crime is the application of local indicators of spatial association (LISA) statistics (Anselin, 1995; Getis and

Ord, 1996). LISA statistics assess the local association between data by comparing local averages to global averages. For this reason they are useful in adding definition to crime hot spots and placing a spatial limit on those areas of highest crime event concentration (Ratcliffe and McCullagh, 1998). One of the more applied LISA statistics on crime point events is the  $G_i^*$  statistic (see Ratcliffe and McCullagh, 1998, for more details).

The  $G_i^*$  statistic is applied to a grid cell output, such as a quartic kernel density estimation map, from which local associations are compared against the global average. The user typically is required to enter a search distance within which local

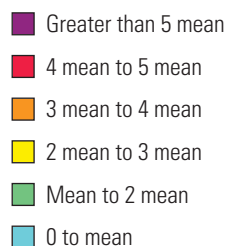
associations are explored and a significance level is related to the confidence in the final output. The search distance usually is set to three times the grid cell size of the original kernel estimation surface (i.e., for data in this chapter, the search distance was set to 90 m). Although levels in significance can often range between 99.9 percent, 99 percent, and 95 percent, this range has far less effect over eventual hot spot areas than parameters set for grid cell size or bandwidth (Ratcliffe, 1999b). The most common significance level to apply is 99.9 percent.

Exhibit 17 shows the result of the  $G_i^*$  LISA statistic (mapped as grey areas) for robbery. This LISA output has been

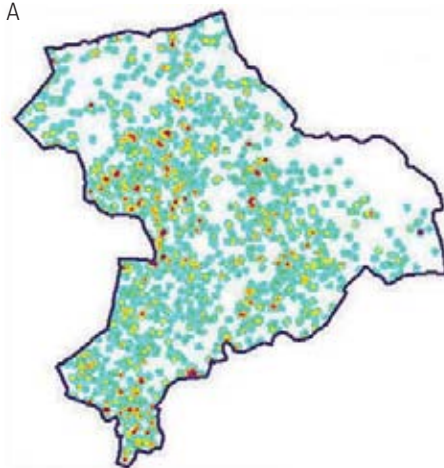
#### Exhibit 15. Crime hot spot thresholds for vehicle crime

**A: Bandwidth 62 m (K2)**

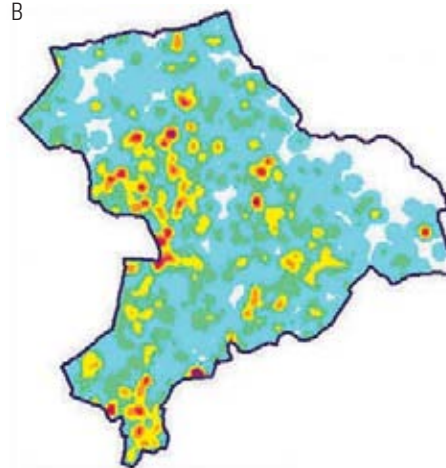
**B: Bandwidth 138 m (K7)**



A



B



mapped with the quartic kernel density estimation surface generated using a bandwidth of 117 m (K2). The map shows the matching between this LISA output and kernel value results above the 3-mean threshold. This LISA output adds definition to the quartic kernel density estimation, indicating the level at which hot spots can be more clearly distinguished from other levels of crime concentration.

## Additional elements to consider

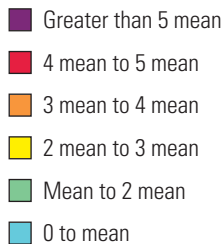
All the methods described above consider volume crime patterns and do not consider any underlying population distribution.

Therefore, in certain cases, such as hot spot maps for residential burglary, the crime hot spots may simply be displaying locations of high housing density. Rate maps can be created to take account of the underlying population distribution. For residential burglary, maps that consider the underlying housing distribution can be generated by calculating a rate of burglaries per certain number of households (e.g., burglaries per 1,000 households). The most reliable housing counts available usually are those sourced from the population census. The household counts usually are collected at the census block level, thus allowing the analysts to generate maps representing the distribution of residential burglary by their rates. However,

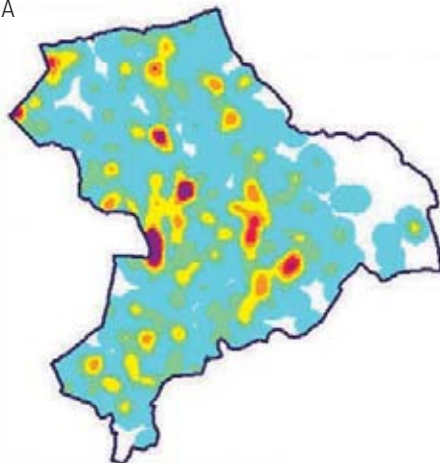
**Exhibit 16. Quartic kernel density estimation hot spot maps for robbery and residential burglary using the incremental mean approach**

**A: Robbery: Bandwidth 207 m (K6)**

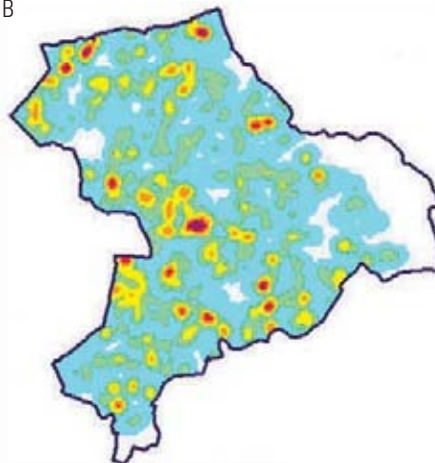
**B: Residential burglary: Bandwidth 161 m (K6)**



A



B



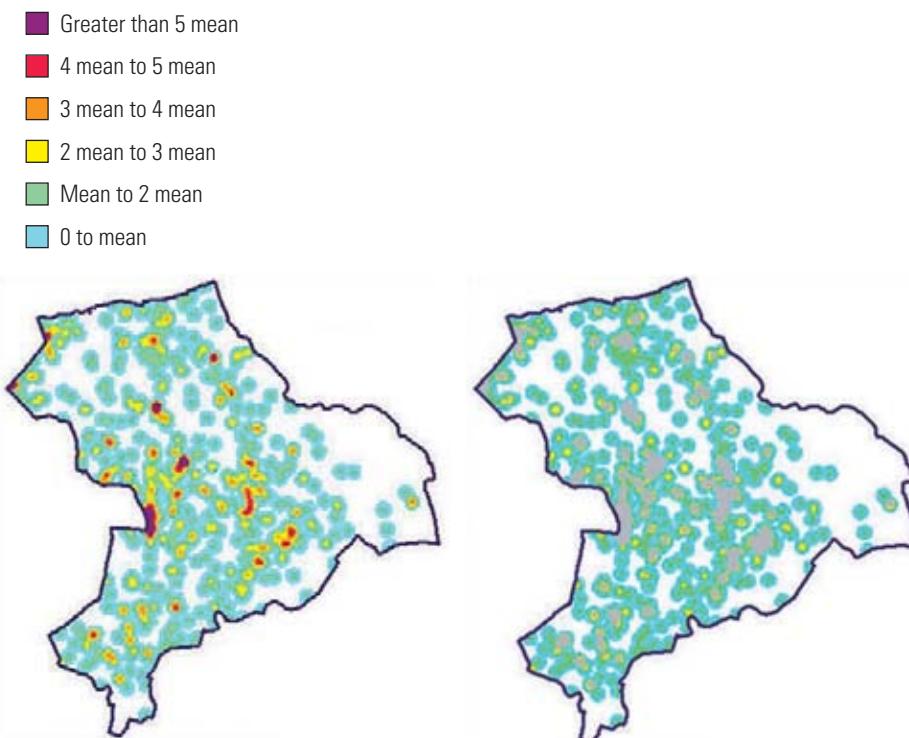
although the underlying spatial distribution of housing is considered, the hot spot maps generated still suffer from the problems described in the section on geographic boundary thematic maps.

Also, suitable denominators for calculating rates rarely are available for other crime types, such as robbery and vehicle crime. Analysts often use population counts as denominators for calculating rates for these other crime types. This approach, however, may merely create hot spot mapping output that misleads by exaggerating the crime problem in town centers that have few residents but a concentration of crimes such as robbery and vehicle crime. Ideally, it is preferable to use denominators that are directly relevant to the crime type for which the analysts wish to create

a rate. In the case of residential burglary, analysts usually have this with census tract household counts.

Suitable denominators for calculating rates for other types of crime are rarely available. For robbery, a suitable denominator for calculating rates would be pedestrian counts for the area; for vehicle crime, a suitable denominator would be vehicle counts. The analysts would also wish to access these counts at a more precise source, such as point sample locations from which estimations at unsampled locations could be interpolated. This would then allow the analysts to be more flexible in the area rate calculation by not restricting them to counts aggregated to geographic boundary areas of varying sizes and shapes.

**Exhibit 17. Robbery quartic kernel density estimation surface (117 m (K2) bandwidth) and Gi\* LISA statistic output (grey areas) derived from this robbery surface**





In the United Kingdom, several local authority and police forces have access to local property gazetteers. These are point-specific geographic records of all properties across the local authority area. Every property across the local authority area has a unique record that can be mapped exactly (within 1 meter) to the property location. These records often also describe property use (e.g., residential or nonresidential). When a local property gazetteer such as this exists, crime analysts can use accurate point-based records of the distribution of residential properties to create continuous surface crime rate hot spot maps that consider both the point-based distribution of crime and the point-based underlying residential property distribution.

## Variations in time

Each hot spot map considered in this chapter accounts only for a specific snapshot period in time. New areas of research are beginning to explore space-time interaction (Ratcliffe and McCullagh, 1998) and the use of test statistics such as those devised by Knox and Mantel (see Bailey and Gatrell, 1995). These methods aim to reveal whether certain types of crime display temporal hot spots in particular areas (e.g., crime hot spots that emerge only on certain days of the week). A step that many crime analysts already have taken in this direction is the creation of crime hot spot animations to visualize space and time interaction. Each frame produced for the animation is itself a hot spot map that is then knitted to a time sequence of other hot spot maps to produce a series that can be displayed as an animation. (For an example of a hot spot map animation, visit <http://www.crimereduction.co.uk/toolkits/fa020405.htm>). Animated hot spot maps are very alluring, but require direct application if they are not to be merely gimmicks of the crime mapping trade.

## Conclusion

The methods and techniques in this chapter have indicated how crime analysts can examine crime-related data to visualize and understand crime hot spots. Preliminary global statistics have shown how simple-to-apply tests can reveal an understanding of what is to be expected in a hot spot map, even before the map has been created. Tests for clustering are particularly important. Analysts may waste valuable resource time in their attempts to create a crime hot spot map if a test such as the NNI quickly reveals that no clusters, and thus no hot spots, exist in their data. The different mapping techniques have revealed the different applications to which they are suited and demonstrated the advantages and disadvantages in their underlying routines and the mapping outputs they generate. The kernel density estimation method in particular has been demonstrated as being more than just a method that presents an attractive map of crime, but is a more robust technique suited to understanding spatial patterns of crime hot spots.

Also important to remember is that map production is an iterative process. The first map produced is very rarely the one presented to the target audience. The intended message should also be seen as the driving force behind what the map should look like. Map creation and design requires flexibility. Methods and techniques described in this chapter retain this flexibility but suggest some simple-to-apply operations that help to understand crime hot spots.

## References

Anselin, L. 1992. *SpaceStat: A Program for the Statistical Analysis of Spatial Data*. Santa Barbara, CA: National Center for Geographic Information and Analysis, University of California.